

COA Imp

Module 1: Computer Evolution & Arithmetic

Includes architecture, arithmetic operations, buses, and register transfers.

- **Von Neumann Architecture**
- **Harvard Architecture**
- **Booth's Algorithm**
- **Register & Register Transfer Mechanism**
- **Bus system (single & multiple bus)**
- **Arbitration Technique**

Questions

- 1. Booth's Algorithm**
- 2. Harvard Architecture**
- 3. Von Neumann Architecture**
- 4. Register**
- 5. Register Transfer Mechanism**
- 6. Arbitration Technique**
- 7. Register Transfer Mechanics**
- 8. Bus System (Single & Multiple)**

Module 2: Control Unit

Focus on control design and micro-operations.

- **Hardwired Control**

- **Microprogrammed Control**
- **Microinstruction & Microprogram**
- **Microinstruction Sequencing**
- **Pipelines & performance**

Questions

- 1. Microprogrammed Control**
 - 2. Hardwired Control Design**
 - 3. Microinstruction Sequencing**
 - 4. Pipelines & Performance**
 - 5. Microinstruction and Microprogram**
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Module 3: Processor Organization & I/O Organization

Includes processor design, memory, and storage.

- **Memory hierarchy**
- **Cache memory & mapping**
- **Auxiliary memory**
- **Stack organization**

Questions

- 1. Memory Hierarchy**
- 2. Auxiliary Memory Devices**
- 3. Magnetic Disk & Tape Mechanism**
- 4. Cache Mapping (Main Memory ↔ Cache)**
- 5. Stack Organization**

Module 4: Interrupts

Includes I/O and communication.

- **Interrupts & types**
- **Modes of data transfer**
- **Programmed I/O**
- **Interrupt-driven I/O**
- **DMA**
- **I/O interface & ports**

Questions

- 1. Interrupts and Types**
 - 2. Programmed I/O & Interrupt-driven I/O**
 - 3. I/O Interface & I/O Ports**
 - 4. Modes of Data Transfer**
 - 5. DMA**
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Extra Question...

Module 1: Computer Evolution & Arithmetic (Missing Questions)

- **Explain computer evolution (generations of computers)**
- **Explain functional units of a computer**
- **Explain bus architecture in detail**
- **Explain arithmetic logic unit (ALU)**
- **Explain shift micro-operations**
- **Perform binary addition and subtraction using 2's complement**

Module 2: Control Unit (Missing Questions)

- **Explain control unit functions**
- **Draw and explain block diagram of control unit**
- **Explain instruction cycle (fetch–decode–execute)**
- **Explain micro-operations in instruction execution**
- **Explain control word format**
- **Difference between horizontal and vertical microprogramming**

Module 3: Processor Organization & I/O (Missing Questions)

- **Explain instruction format**
- **Explain addressing modes**
- **Explain general register organization (with diagram)**
- **Explain RISC vs CISC architecture**

- Explain RAM and ROM types
- Explain virtual memory
- Explain cache performance (hit ratio, miss penalty)

Module 4: Interrupts (Missing Questions)

- Explain interrupt cycle
 - Explain priority interrupts
 - Explain daisy chaining technique
 - Explain I/O channels and processors
 - Explain synchronous vs asynchronous communication
 - Explain standard communication interfaces (like USB, etc.)
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